NATIONAL UNIVERSITY OF MODERN LANGUAGES

Lab no 3

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Lab Report:

Software Quality Management Lab Report

Case Study: Development of an Online Banking System

This report analyzes the quality management activities involved in the development of an Online Banking System. The project requires high security, reliability, performance, and compliance with banking regulations. Based on the case study, quality-related activities were categorized into three major areas: Quality Planning, Software Quality Assurance (SQA), and Software Quality Control (SQC).

Quality Planning focuses on defining standards, processes, and responsibilities before development begins. Software Quality Assurance ensures that the development team follows defined procedures and standards during the project. Software Quality Control focuses on identifying defects in the product through reviews and testing activities. The following table summarizes the activities identified from the case study.

Activity from Case Study	SQA	SQC	Quality Plan	Justification
Quality Management Plan prepared (ISO standards, roles, documentation guidelines)			✓	Defines overall quality framework, standards, responsibilities, and procedures.
Definition of quality objectives (security compliance, 99.9% availability, fast processing)			✓	Establishes measurable quality goals for the system.
Coding standards, naming conventions, and documentation templates established	✓			Ensures development follows standardized quality practices.
Risk register created (data breach, downtime, integration risks)	✓			Identifies and manages risks affecting software quality.
Requirement review meetings before development	✓			Ensures requirements are clear, complete, and feasible.
Scrum master enforcing Agile practices (stand-ups, sprint planning, retrospectives)	✓			Ensures proper process compliance and project quality management.
Internal audits for documentation and coding guidelines	✓			Verifies adherence to defined quality standards and procedures.
Peer code reviews before merging into main branch	✓			Prevents defects early and improves code quality.
Configuration management and	✓			Maintains controlled

version control				development environment and traceability.
Unit testing of modules (login, transactions)		✓		Detects defects in individual software components.
Integration testing between database, gateway, and UI		✓		Ensures modules interact correctly.
System testing in staging environment		✓		Validates complete system functionality.
Security testing (penetration testing, vulnerability scanning)		✓		Ensures system protection against cyber threats.
Performance testing for concurrent users		✓		Verifies system performance under heavy load.
Automated regression testing before sprint releases		✓		Ensures new updates do not break existing features.
Defect logging, prioritization, and retesting		✓		Tracks and resolves identified software defects.
User Acceptance Testing (UAT) with bank staff and customers		✓		Confirms system meets real business requirements.
Post-deployment monitoring and post-release reviews	✓			Helps improve processes and prevent future quality issues.

Conclusion

The implementation of proper Quality Planning, SQA, and SQC activities ensures that the Online Banking System maintains high standards of reliability, security, and performance. Planning defines the quality framework, assurance ensures the correct processes are followed, and control activities help detect and fix defects in the product. Together, these practices support the successful delivery of a secure and efficient banking system.